Lecture plan · for the lecturer

A stack of neurons

University, 1

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- One neuron, one cut
- A stack can bend

Activity: A one-phone seminar

1. Vote (5 min): Show the points in a ring; pairs vote on whether one straight line can split them, and say why.
2. By hand (10 min): Each pair computes one forward step on paper: two inputs, a hidden layer of two ReLU neurons, one output. Then they say what the output would be without the activation.
3. Playground (15 min): One phone between two, on the Circle data: train with 1 hidden layer and Activation “None (linear)”, then with tanh, and say which change bent the boundary.
4. Mark (10 min): Each student writes one sentence on what the activation does. Mark the sentence, not a screenshot of a loss going down.
5. Exit (5 min): The exit ticket: name the part that bends the stack.

Discussion

- Why is depth not the same as “more”?

Exit ticket

- Name the part that bends the stack.

Try it in your next lecture
Set the seminar: pairs vote on a boundary a line misses before anyone opens the playground.

Seminar sheet · for students

A stack of neurons

University, 1

Name _____ Group _____ Date _____

1 **Without an activation, a stack of layers becomes…**

◯ One straight cut ◯ Two separate networks ◯ A curve around the points

2 **A hidden layer with a nonlinear activation can separate shapes one neuron cannot.**

◯ True ◯ False

3 **Match the part to its job.**

Weight	• • • •	A bend, so the stack is not one big line
Bias	• • • •	How much an input counts
Activation	• • • •	A shift before the activation

4 **Put a forward step in order.**

☐ Apply the activation

☐ Multiply inputs by weights

☐ Add the bias

5 **Inputs x = (1, 2). A hidden ReLU neuron has weights (0.5, −1) and bias 1. What does it output?**

◯ -0.5 ◯ 0 ◯ -1.5

Answers · for the lecturer

A stack of neurons

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1 **One straight cut**

Without the bend, stacked linear layers multiply out to one linear layer: one straight cut.

2 **True**

The playground shows a ring that one line misses and a small network learns.

3 **Weight — How much an input counts**
Bias — A shift before the activation
Activation — A bend, so the stack is not one big line

Without an activation, stacked linear layers collapse into one linear layer.

4 **Top to bottom: 3, 1, 2**

The activation is last. It bends the sum.

5 **0**

0.5 × 1 − 1 × 2 + 1 = −0.5, and ReLU turns a negative sum into 0.

For you to think about

Which sentence will you mark: the one about depth, or the one about the activation?